Unit: 6 Unit 2: Exponents Radicals **Name:** ______________
Topic: Rationalizing Denominator **Date:** __________

Advanced (Carolina Reaper)

Q1. Rationalize the denominator of $\frac{1}{\sqrt{2}}$.

(A) $\frac{\sqrt{2}}{2}$
(B) $\frac{1}{2\sqrt{2}}$
(C) $\frac{2}{\sqrt{2}}$
(D) $\frac{\sqrt{2}}{\sqrt{2}}$
(E) $\frac{2}{2\sqrt{2}}$

Q2. Simplify $\frac{1}{\sqrt{5}+\sqrt{3}}$ by rationalizing the denominator.

(A) $\frac{\sqrt{5}-\sqrt{3}}{2}$
(B) $\frac{\sqrt{5}+\sqrt{3}}{2}$
(C) $\frac{\sqrt{15}-3}{2}$
(D) $\frac{\sqrt{15}+3}{2}$
(E) $\frac{\sqrt{15}-3}{-2}$

Q3. Rationalize the denominator of $\frac{2}{\sqrt{6}-\sqrt{2}}$.

(A) $\frac{2(\sqrt{6}+\sqrt{2})}{4}$
(B) $\frac{2(\sqrt{6}-\sqrt{2})}{4}$
(C) $\frac{\sqrt{6}+\sqrt{2}}{2}$
(D) $\frac{2\sqrt{6}+2\sqrt{2}}{4}$
(E) $\frac{\sqrt{6}-\sqrt{2}}{2}$

Q4. Rationalize the denominator of $\frac{3}{\sqrt{10}+\sqrt{2}}$.

(A) $\frac{3(\sqrt{10}-\sqrt{2})}{8}$
(B) $\frac{3(\sqrt{10}+\sqrt{2})}{8}$
(C) $\frac{3\sqrt{10}-3\sqrt{2}}{8}$
(D) $\frac{3\sqrt{10}+3\sqrt{2}}{8}$
(E) $\frac{\sqrt{10}-\sqrt{2}}{8}$

Q5. Simplify $\frac{1}{\sqrt{7}-\sqrt{3}}$ by rationalizing the denominator.

(A) $\frac{\sqrt{7}+\sqrt{3}}{4}$
(B) $\frac{\sqrt{7}-\sqrt{3}}{4}$
(C) $\frac{\sqrt{21}+3}{4}$
(D) $\frac{\sqrt{21}-3}{4}$
(E) $\frac{\sqrt{21}-3}{-4}$

Q6. Rationalize the denominator of $\frac{2}{\sqrt{3}+\sqrt{5}}$.

(A) $\frac{2(\sqrt{3}-\sqrt{5})}{-2}$
(B) $\frac{2(\sqrt{3}+\sqrt{5})}{-2}$
(C) $\frac{2\sqrt{3}-2\sqrt{5}}{-2}$
(D) $\frac{2\sqrt{3}+2\sqrt{5}}{-2}$
(E) $\frac{2(\sqrt{3}-\sqrt{5})}{2}$

Q7. Simplify $\frac{1}{\sqrt{2}+\sqrt{6}}$ by rationalizing the denominator.

(A) $\frac{\sqrt{2}-\sqrt{6}}{-4}$
(B) $\frac{\sqrt{2}+\sqrt{6}}{-4}$
(C) $\frac{\sqrt{2}-\sqrt{6}}{4}$
(D) $\frac{\sqrt{2}+\sqrt{6}}{4}$
(E) $\frac{\sqrt{12}-\sqrt{2}}{-4}$

Q8. Rationalize the denominator of $\frac{3}{\sqrt{5}-\sqrt{2}}$.

(A) $\frac{3(\sqrt{5}+\sqrt{2})}{3}$
(B) $\frac{3(\sqrt{5}+\sqrt{2})}{5-2}$
(C) $\frac{3\sqrt{5}+3\sqrt{2}}{3}$
(D) $\frac{3(\sqrt{5}-\sqrt{2})}{3}$
(E) $\frac{\sqrt{5}+\sqrt{2}}{3}$

Open Ended Questions (FRQ)

Q9. Rationalize the denominator of $\frac{2}{\sqrt{6}+\sqrt{2}}$ and simplify.

Q10. Simplify $\frac{1}{\sqrt{10}-\sqrt{2}}$ by rationalizing the denominator and explain each step.

Chef's Tips

- Remind students to rationalize the denominator by multiplying the numerator and denominator by the conjugate of the denominator.
- Emphasize that rationalizing the denominator is a crucial step in simplifying expressions with square roots.
- Encourage students to show all their work and explain each step of the problem-solving process.